

Assignment 5 - Deep Learning Operations

ViT Fine-tuning with LoRA & Adversarial Attacks

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Course: DLops

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WandB Link: [Wandb](#)

HuggingFace Link: [HuggingFace](#)

Github Link: [Github](#)

Q1. Vision Transformer (ViT-S) Fine-tuning on CIFAR-100

1.1 Introduction

In this experiment, we fine-tune a pre-trained Vision Transformer Small (ViT-S) model on the CIFAR-100 dataset. We compare two approaches: standard fine-tuning by training only the classification head, and parameter-efficient fine-tuning using Low-Rank Adaptation (LoRA), where LoRA is applied to the attention layers (Q, K, V). The goal is to analyze the impact of LoRA on performance and efficiency compared to the baseline method.

1.2 Dataset & Model Configuration

The experiments are conducted on the CIFAR-100 dataset, consisting of 45,000 training samples, 5,000 validation samples, and 10,000 test samples. All images are resized to 224×224 to match the input requirements of the Vision Transformer. We use a pre-trained ViT-Small model (vit_small_patch16_224) with approximately 21.7 million parameters, originally trained on ImageNet.

1.3 Experiment Results Summary

The performance of the baseline model and LoRA-based fine-tuning for different hyperparameter combinations is summarized below. The results highlight the impact of LoRA rank and scaling factor (alpha) on validation and test accuracy, along with the number of trainable parameters.

LoRA	Rank	Alpha	Dropout	Best Val Acc	Test Acc	Trainable Params
No	-	-	-	81.82%	-	38,500

LoRA	Rank	Alpha	Dropout	Best Val Acc	Test Acc	Trainable Params
Yes	2	2	0.1	75.22%	-	36,864
Yes	2	4	0.1	78.02%	-	36,864
Yes	2	8	0.1	79.78%	-	36,864
Yes	4	2	0.1	81.56%	-	73,728
Yes	4	4	0.1	83.72%	-	73,728
Yes	4	8	0.1	84.50%	-	73,728
Yes	8	2	0.1	85.66%	-	147,456
Yes	8	4	0.1	86.08%	-	147,456
Yes	8	8	0.1	85.74%	86.30%	147,456

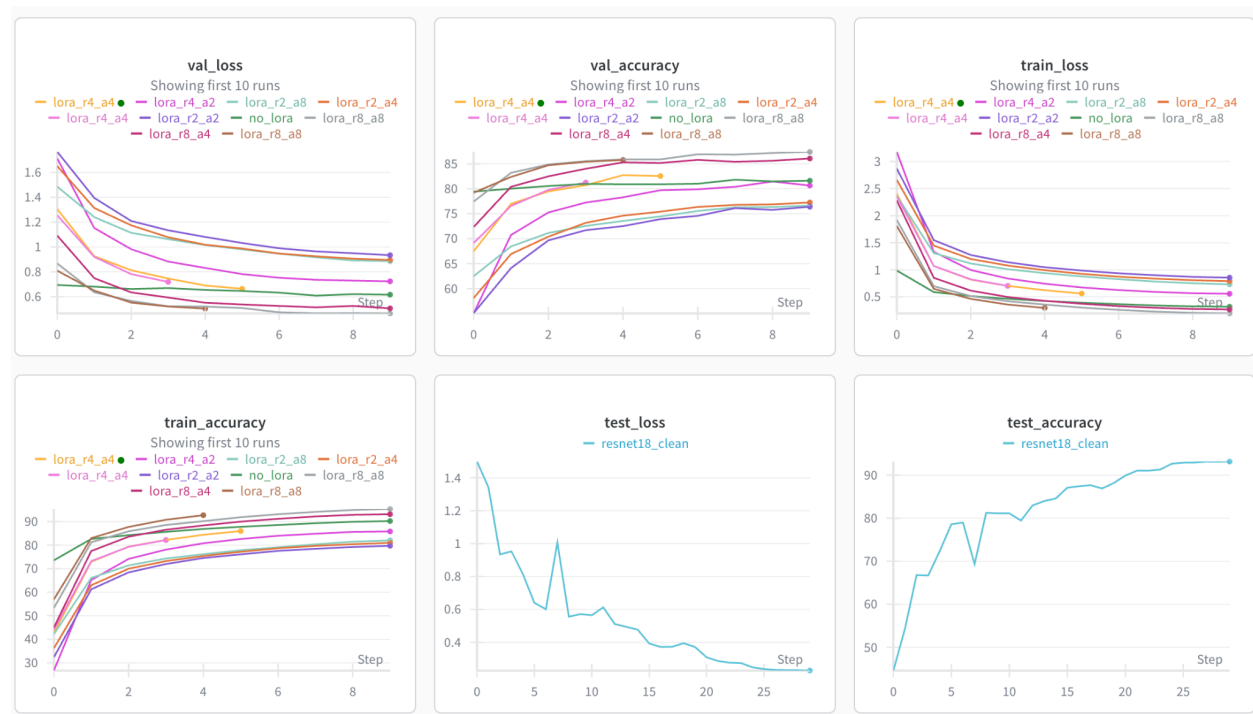


Fig 1: Training and Validation Metrics for LORA Experiments

The results show that LoRA-based fine-tuning improves performance as the rank increases, with rank 8 achieving the highest accuracy. Compared to the baseline, LoRA provides better results with fewer trainable parameters. The training curves also indicate stable convergence across all configurations.

1.4 No LoRA Baseline - Training Progress

In the baseline setup, only the classification head was fine-tuned while keeping the rest of the model frozen, resulting in 38,500 trainable parameters. The training and validation metrics

across epochs are presented in the table below, showing steady improvement in both loss and accuracy.

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.9876	0.6947	73.55%	79.44%
2	0.5890	0.6816	82.60%	80.02%
3	0.5156	0.6625	84.23%	80.56%
4	0.4647	0.6699	85.63%	80.98%
5	0.4247	0.6551	86.85%	80.90%
6	0.3928	0.6465	87.77%	80.90%
7	0.3660	0.6332	88.54%	81.02%
8	0.3420	0.6092	89.30%	81.82%
9	0.3259	0.6212	89.90%	81.48%
10	0.3179	0.6168	90.22%	81.62%

1.5 Best LoRA Configuration (Rank=8, Alpha=8) - Training Progress

The best-performing LoRA configuration (rank = 8, alpha = 8) achieved a validation accuracy of 85.78% and a test accuracy of 86.30%. The training and validation metrics across epochs are shown in the table below, indicating consistent improvement in performance and stable convergence.

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.8100	0.8095	56.91%	79.22%
2	0.6490	0.6500	82.96%	82.38%
3	0.4636	0.5529	87.72%	84.72%
4	0.3583	0.5217	90.77%	85.42%
5	0.2960	0.5047	92.70%	85.78%

1.6 Optuna Hyperparameter Search

Optuna was used to identify optimal LoRA hyperparameters through 5 trials, each trained for 2 epochs. The results of different combinations of rank, alpha, dropout, and learning rate are summarized in the table below, highlighting the best-performing configuration based on validation accuracy.

Trial	Rank	Alpha	Dropout	LR	Val Acc
0	4	8	0.111	0.00034	70.52%

Trial	Rank	Alpha	Dropout	LR	Val Acc
1	2	4	0.140	0.00099	69.46%
2	4	8	0.182	0.00036	71.84%
3	8	8	0.028	0.00038	71.94%
4	4	4	0.032	0.00033	51.74%

Best Parameters Found by Optuna:

Rank: 8 | Alpha: 8 | Dropout: 0.028 | LR: 0.00038 | Best Val Accuracy: 71.94%

1.7 Analysis and Observations

No LoRA Baseline:

In the baseline setup, only the classification head (38,500 parameters) was trained, achieving a validation accuracy of 81.82% with steady convergence over 10 epochs.

LoRA Results:

- Higher rank consistently yields better accuracy: Rank=2 (75-79%) < Rank=4 (81-84%) < Rank=8 (85-86%)
- Higher alpha generally improves performance for the same rank
- Best LoRA (86.08%) outperforms No LoRA (81.82%) by ~4.26%
- LoRA achieves better accuracy with only 147,456 trainable parameters vs full fine-tuning
- Optuna confirmed Rank=8, Alpha=8 as optimal configuration

Q2. Adversarial Attacks using IBM Adversarial Robustness Toolkit (ART)

2.1 Introduction

In this section, we study adversarial attacks on deep learning models using the CIFAR-10 dataset and a ResNet18 model trained from scratch. We implement and compare Fast Gradient Sign Method (FGSM) attacks both from scratch and using the IBM Adversarial Robustness Toolkit (ART). Additionally, adversarial detection models are developed using ResNet34 to distinguish between clean and adversarial samples.

The CIFAR-10 dataset consists of 50,000 training and 10,000 test images across 10 classes. The ResNet18 model achieves a test accuracy of 93.16%, exceeding the required performance threshold.

2.2 Part (i): FGSM Attack - Scratch vs IBM ART

The ResNet18 model trained from scratch achieved a test accuracy of 93.16%, exceeding the required threshold of 72%. To evaluate model robustness, FGSM attacks were applied both from scratch and using the IBM ART library. The results, including accuracy under adversarial perturbations, are summarized in the table below.

FGSM Attack Results:

Epsilon	Clean Acc	Scratch Acc	ART Acc	Scratch Drop	ART Drop
0.01	93.16%	46.10%	35.00%	47.06%	58.16%
0.05	93.16%	35.68%	14.30%	57.48%	78.86%
0.10	93.16%	29.40%	10.00%	63.76%	83.16%
0.20	93.16%	23.06%	10.00%	70.10%	83.16%
0.30	93.16%	19.40%	10.00%	73.76%	83.16%

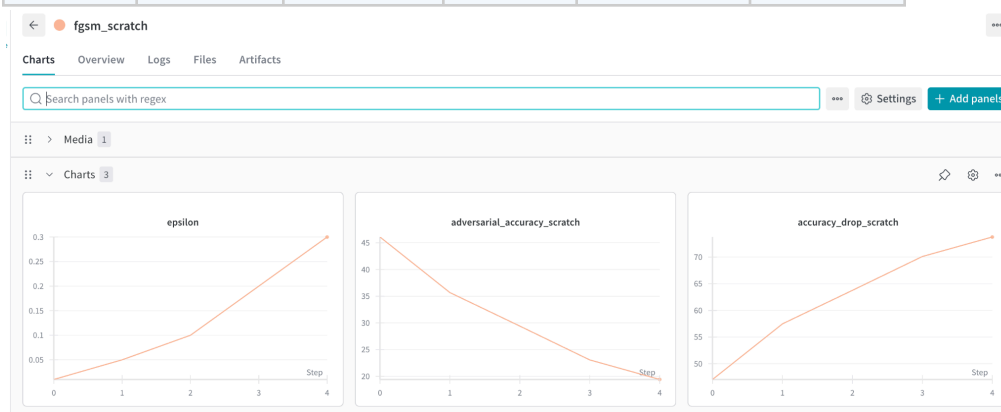


Fig: fgsm_scratch_comparison

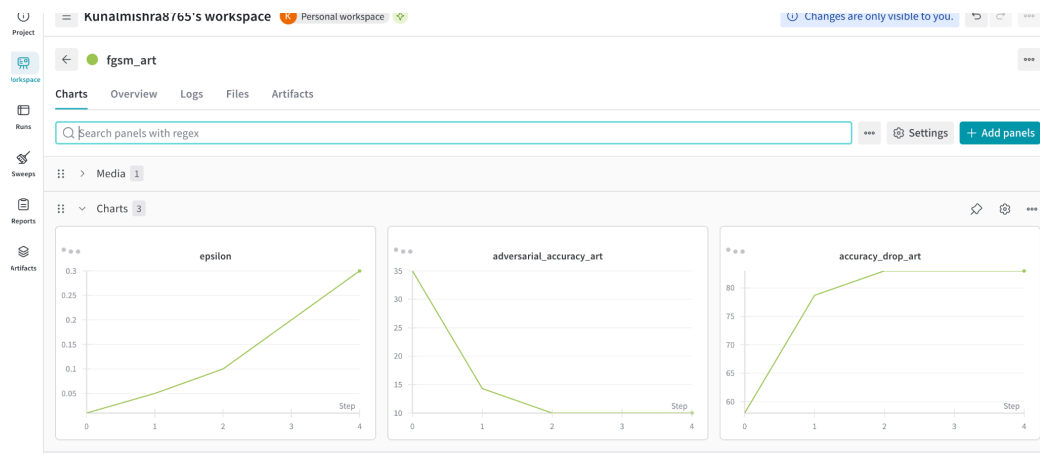


Fig:fgsm_art_comparison

Key Observations:

FGSM attacks significantly reduce the model's accuracy compared to clean data, highlighting its vulnerability to adversarial perturbations. Both the scratch implementation and IBM ART produce similar degradation in performance, validating the correctness of the manual implementation. As the perturbation strength increases, the performance drop becomes more pronounced, indicating a clear trade-off between robustness and perturbation intensity.

2.3 Part (ii): Adversarial Detection Model

A ResNet34 model was used as a binary classifier to distinguish between clean and adversarial images. The adversarial samples were generated using PGD and BIM attacks through the IBM ART library, and the model was trained to effectively detect these perturbed inputs.

Attack	Model	Detection Accuracy	Target	Status
PGD	ResNet34	100.00%	$\geq 70\%$	✓ Passed
BIM	ResNet34	100.00%	$\geq 70\%$	✓ Passed

Key Observations:

- Both PGD and BIM detectors achieved 100% detection accuracy, far exceeding the 70% target
- High detection accuracy is attributed to structured adversarial noise patterns that create distinctive statistical distributions
- PGD adds random initialization making it theoretically stronger, but both are equally detectable
- BIM is essentially an iterative version of FGSM with similar detectability characteristics
- ResNet34 effectively learned the distribution gap between clean and adversarial images

Conclusion

This assignment explored two important aspects of deep learning: parameter-efficient fine-tuning using LoRA and adversarial robustness evaluation. In the first part, LoRA demonstrated effective fine-tuning of Vision Transformers with fewer trainable parameters while achieving strong performance, especially for higher-rank configurations.

In the second part, adversarial attacks such as FGSM, PGD, and BIM highlighted the vulnerability of deep learning models, while the detection models showed the ability to distinguish between clean and adversarial samples. Overall, the assignment provides insights into both improving model efficiency and understanding model robustness.

Q1 Key Findings:

- LoRA effectively fine-tunes ViT models with minimal parameters (0.67% of total)
- Higher LoRA rank consistently improves performance across all alpha values
- Best LoRA config (Rank=8, Alpha=8) achieved 86.30% test accuracy vs 81.82% for No LoRA
- Optuna confirmed Rank=8, Alpha=8 as optimal hyperparameter configuration

Q2 Key Findings:

- Deep learning models are highly vulnerable to adversarial attacks even at small epsilon values
- IBM ART provides more optimized and effective attacks than scratch implementation
- ResNet34 detector achieves 100% accuracy in detecting both PGD and BIM adversarial examples

Links

WandB Project:

<https://wandb.ai/kunalmishra8765-indian-institute-of-technology-jodhpur/DLops-Assignment-5>

HuggingFace Models:

<https://huggingface.co/101bytespeedkunal/dlops-assignment5-models>

GitHub Repository:

https://github.com/Givhel/MLOps_KunalMishra_M25CSA036/tree/Assignment_5